2.23 Differentiation: Connected Rates of Change

Worked Solutions 1–16

1 a)

$A = \pi r^2$

1 b)

$A = \frac{1}{2}x^2 \sin 60° = \frac{\sqrt{3}}{4}x^2$

1 c)

$V = \pi r^2 h$

1 d)

$V = \frac{4}{3}\pi r^3$

1 e)

$A = 4\pi r^2$

1 f)

$V = \frac{1}{3}\pi r^2 h$

2 a)

$$\frac{dA}{dr} = \pi r, \qquad \frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} = \pi r \times 2 = 2\pi r$$

2 b)

$$\frac{dy}{dx} = 20x, \qquad \frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = 20x \times -0.5 = -10x$$

2 c)

$$\frac{dP}{dx} = 6x^2 - 1, \qquad \frac{dP}{dt} = \frac{dP}{dx} \times \frac{dx}{dt} = 3(6x^2 - 1) = 18x^2 - 3$$

2 d)

$$\frac{dM}{dx} = -10x^4 + 2x$$

$$\frac{dM}{dt} = \frac{dM}{dx} \times \frac{dx}{dt} = \frac{1}{2}(-10x^4 + 2x) = -5x^4 + x$$

3)

$$\frac{dx}{dt} = 10, \qquad A = x^2, \qquad \frac{dA}{dx} = 2x$$
$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt} = 2x \times 10 = 20x$$
$$\left.\frac{dA}{dt}\right|_{x=4} = 80 \text{ cm}^2\,\text{s}^{-1}$$

4)

$$\frac{dr}{dt} = 3, \qquad A = \pi r^2, \qquad \frac{dA}{dr} = 2\pi r$$
$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} = 2\pi r \times 3 = 6\pi r$$
$$\left.\frac{dA}{dt}\right|_{r=5} = 30\pi \text{ cm}^2\,\text{s}^{-1}$$

5)

$$\frac{dr}{dt} = -0.1$$

Perimeter: $c = 2\pi r \Rightarrow \frac{dc}{dr} = 2\pi$

$$\frac{dc}{dt} = \frac{dc}{dr} \times \frac{dr}{dt} = 2\pi \times -0.1 = -0.2\pi$$

Rate of decrease: $\frac{dc}{dt} = 0.2\pi$

6)

$$\frac{dx}{dt} = 12, \qquad A = \tfrac{1}{2}x^2 \sin 60^\circ = \tfrac{\sqrt{3}}{4}x^2, \qquad \frac{dA}{dx} = \tfrac{\sqrt{3}}{2}x$$
$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt} = \tfrac{\sqrt{3}}{2}x \times 12 = 6\sqrt{3}x$$
$$\left.\frac{dA}{dt}\right|_{x=2\sqrt{3}} = 6\sqrt{3} \times 2\sqrt{3} = 36 \text{ cm}^2\,\text{s}^{-1}$$

7)

$$\frac{dA}{dt} = 12\pi, \qquad A = 4\pi r^2, \qquad \frac{dA}{dr} = 8\pi r$$
$$\frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt} = \frac{1}{8\pi r} \times 12\pi = \frac{3}{2r}$$
$$\left.\frac{dr}{dt}\right|_{r=3} = \frac{3}{6} = 0.5 \text{ cm}\,\text{s}^{-1}$$

8 a)

$V = x^3$, $A = 6x^2$

8 b)

$$\frac{dV}{dt} = 5, \qquad \frac{dV}{dx} = 3x^2, \qquad \frac{dA}{dx} = 12x$$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dV} \times \frac{dV}{dt} = 12x \times \frac{1}{3x^2} \times 5 = \frac{20}{x}$$

8 c)

$\left.\frac{dA}{dt}\right|_{x=4} = \frac{20}{4} = 5 \text{ cm}^2\,\text{s}^{-1}$

9)

$$p = \frac{k}{x} \text{ gives } 1.2 = \frac{k}{50} \text{ so } k = 60$$

So $p = \frac{60}{x}$

$$p = 60x^{-1}, \qquad \frac{dp}{dx} = -60x^{-2}, \qquad \frac{dx}{dt} = 5$$

$$\frac{dp}{dt} = \frac{dp}{dx} \times \frac{dx}{dt} = -60x^{-2} \times 5 = -\frac{300}{x^2}$$

$$\left.\frac{dp}{dt}\right|_{x=25} = -\frac{300}{25^2} = -0.48 \text{ atmospheric units per second.}$$

10)

$$V = \tfrac{4}{3}\pi r^3, \qquad A = 4\pi r^2$$

$$\frac{dV}{dt} = 4, \qquad \frac{dV}{dr} = 4\pi r^2, \qquad \frac{dA}{dr} = 8\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dV} \times \frac{dV}{dt} = 8\pi r \times \frac{1}{4\pi r^2} \times 4 = \frac{8}{r}$$

$$\left.\frac{dA}{dt}\right|_{r=25} = \frac{8}{25} = 0.32 \text{ cm}^2\,\text{s}^{-1}$$

11 a)

$$V = \pi r^2 h, \qquad \frac{dV}{dt} = 10, \qquad \frac{dV}{dh} = \pi r^2$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{1}{\pi r^2} \times 10 = \frac{10}{\pi r^2}$$

11 b)

$\left.\frac{dh}{dt}\right|_{r=45} = \frac{10}{\pi \times 45^2} = 0.00157 \text{ cm}\,\text{s}^{-1}$

12)

$$V = \pi r^2 h, \qquad \frac{dV}{dt} = -5, \qquad \frac{dV}{dh} = \pi r^2$$
$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{1}{\pi r^2} \times -5 = -\frac{5}{\pi r^2}$$
$$\left.\frac{dh}{dt}\right|_{r=10} = -\frac{5}{\pi \times 100} = -0.0159 \text{ cm per minute}$$

13 a)

$$\tan 30^\circ = \frac{r}{h} \quad \therefore r = \frac{h}{\sqrt{3}}$$
$$V = \tfrac{1}{3}\pi r^2 h = \tfrac{1}{3}\pi \left(\frac{h}{\sqrt{3}}\right)^2 h = \tfrac{1}{9}\pi h^3 \text{ as req.}$$

13 b)

$\frac{dV}{dh} = \frac{1}{3}\pi h^2$

13 c)

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{3}{\pi h^2} \times 6 = \frac{18}{\pi h^2}$$
$$\left.\frac{dh}{dt}\right|_{h=10} = 0.0573 \text{ cm s}^{-1}$$

14)

$$\frac{dV}{dt} = 3$$

$V = \frac{1}{3}\pi r^2 h$ but since $h = 2r$

$$V = \tfrac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \tfrac{\pi}{12}h^3, \qquad \frac{dV}{dh} = \tfrac{\pi}{4}h^2$$
$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{1}{\frac{\pi}{4}h^2} \times 3 = \frac{12}{\pi h^2}$$
$$\left.\frac{dh}{dt}\right|_{h=4} = \frac{12}{\pi \times 4^2} = \frac{3}{4\pi} = 0.239 \text{ cm s}^{-1}$$

15)

$$V = x^3, \qquad \frac{dV}{dt} = k, \qquad \frac{dV}{dx} = 3x^2$$
$$\frac{dx}{dt} = \frac{dx}{dV} \times \frac{dV}{dt} = \frac{1}{3x^2} \times k = \frac{k}{3} \times \frac{1}{x^2}$$

Therefore inversely proportional to the area of one side of the cube.

16 a)

$$\frac{dr}{dt} = \frac{k}{r^2}$$

$$\frac{1}{2} = \frac{k}{2^2} \Rightarrow k = 2 \text{ so } \frac{dr}{dt} = \frac{2}{r^2}$$

16 b)

$$A = \pi r^2, \qquad \frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} = 2\pi r \times \frac{2}{r^2} = \frac{4\pi}{r}$$

$$r = \sqrt{\frac{A}{\pi}} \quad (r > 0)$$

$$\frac{dA}{dt} = \frac{4\pi}{\sqrt{A/\pi}} = 4\pi \times \sqrt{\frac{\pi}{A}} = \sqrt{\frac{16\pi^3}{A}} \text{ as req.}$$